PLIABILITY, STRONG PLIABILITY AND FREE MULTIEXPONENTIAL OPENNESS ARE EQUIVALENT IN CARNOT GROUPS

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Allowed standard prerequisites (editorial).

- Carnot groups, endpoint maps and standard geometric control theory. Original complete bibliography and source-local definitions retained.

1. Introduction

Consider a Carnot group $\mathbb{G}$ with stratified Lie algebra $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_s$ (see Definition 2.1). The endpoint map $E\colon L^\infty([0,1], \mathfrak{g}_1) \to \mathbb{G}$ (see Definition 2.2) plays a crucial role in studying metric and topological properties of $\mathbb{G}$. The goal of this paper is to explore and clarify the connections among several openness and submersivity properties of the endpoint map and of the multiexponential map

$$\Gamma^{(p)}\colon\ (\mathfrak{g}_1)^p \to \mathbb{G}, \qquad \Gamma^{(p)}(Y_1, \dots, Y_p) := \exp(Y_1) \cdots \exp(Y_p),$$

where $p \in \mathbb{N}$ and $\exp\colon \mathfrak{g}_1 \to \mathbb{G}$ is the exponential map, i.e., the restriction of E to $\mathfrak{g}_1$, seen as the subspace of $L^\infty([0,1], \mathfrak{g}_1)$ made of constant functions. In particular, given a vector $X \in \mathfrak{g}_1$, we are interested in the following properties of the multiexponential map and of $E_X(\cdot) := E(X + \cdot)$, the endpoint map based at X:

- (H) there exists $p \in \mathbb{N}$ such that the map $\Gamma^{(p)}$ is open at $(X, \dots, X) \in (\mathfrak{g}_1)^p$;
- (SbH) there exists $p \in \mathbb{N}$ such that the map $\Gamma^{(p)}$ is a submersion at $(X, \dots, X) \in (\mathfrak{g}_1)^p$;
- (P) the map E is open at X (i.e., E_X is open at 0).

One of the earliest appearances of these conditions in the literature is the observation by Cheeger and Kleiner in [3] that in the three-dimensional Heisenberg group every $X \in \mathfrak{g}_1$ satisfies condition (SbH) with $p = 2$. As a consequence, they deduced the following *cone property*: if a horizontally convex set C contains a point p in its closure and an open ball B, then it also contains the cone over B with vertex at p.

In the paper [2], Arena, Caruso, and Monti generalized the submersivity of the map $(Y_1, Y_2) \mapsto \exp(Y_2)\exp(Y_1)$ to the case where $\mathbb{G}$ is of step two of Métivier type and Morbidelli proved in [8] that the cone property holds for all Carnot groups of step two.

Montanari and Morbidelli later observed in [7] that the cone property can be obtained under the weaker assumption that every $X \in \mathfrak{g}_1$ satisfies condition (H) (which is clearly implied by (SbH)). In addition, they showed that if a vector $X \in \mathfrak{g}_1$ satisfies condition (SbH), then the Carnot–Carathéodory distance from the unit element of $\mathbb{G}$ is Pansu differentiable at $\exp(X)$. (They also showed that (SbH) implies the related notion of *deformable direction* introduced by Pinamonti and Speight in [9]).

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Condition (H) appears again in the paper [10] by Rigot and it is used to characterize precisely monotone subsets of $\mathbb{G}$. We adapt the terminology introduced in [10], by distinguishing between the (H)-*condition* and the *submersive* (H)-*condition* (SbH) introduced above.

On the other hand, condition (P), called *pliability*, was introduced in [4], where Juillet and the second author proved the C^1 Whitney extension property for horizontal curves in Carnot groups where every vector in $\mathfrak{g}_1$ satisfies the property (P) (see also the recent preprint [12] by Speight and Zimmerman for a generalization to horizontal curves tangent to pliable directions). The definition of pliability proposed in [4] is actually given in terms of the openness of the restriction of the endpoint map to a suitable proper subspace of $L^\infty([0,1],\mathfrak{g}_1)$. The equivalence between (P) and pliability as defined in [4] is one of the contributions of this paper. For a precise discussion on this point, see Remark 3.3.

The first question addressed in this paper concerns the relationship between pliability and the (H)-conditions. Later, Sacchelli and the second author introduced in [11] the notion of *strong pliability* (SP), and they improved the results of [4] for horizontal curves in sub-Riemannian manifolds where every vector in $\mathfrak{g}_1$ of the tangent Carnot group satisfies (SP). Condition (SP) and the analogous notion of *p-free* (H)-*condition* (FH) are defined as follows:

(SP) for all $\eta > 0$ there exists $Y \in L^\infty([0,1],\mathfrak{g}_1)$ such that $\|Y\|_\infty < \eta$, $E_X(Y) = E_X(0)$, and the map E_X is a submersion at Y;

(FH) for all $\eta > 0$ there exist an integer $p \geq 2$ and $W_1, \dots, W_p \in \mathfrak{g}_1$ such that $\|W_i - X\| < \eta$ for $i = 1, \dots, p$, $\Gamma^{(p)}(W_1, \dots, W_p) = \Gamma^{(p)}(X, \dots, X)$, and $\Gamma^{(p)}$ is a submersion at $(W_1, \dots, W_p)$.

Finally, note that (SP) holds true in particular for $X \in \mathfrak{g}_1$ that are regular points of the endpoint map (i.e., those X for which the straight line $t \mapsto \exp(tX)$ is not abnormal). This property motivates a last condition:

(Reg) the map E is a submersion at X (i.e., E_X is a submersion at 0).

Our main result is the following.

Theorem 1.1. *The following hold:*

- *pliability, strong pliability, and the* (FH)*-condition are equivalent;*
- *regularity* (Reg) *and the submersive* (H)*-condition are equivalent and imply the* (H)*-condition, which, in turn, implies the preceding three conditions.*

We note that the equivalence between the (H)-condition and the submersive (H)-condition does not hold; see Remark 3.13. Theorem 1.1 is proved in Section 3 in a slightly more general version, see Theorem 3.5.

2. Preliminaries

Let us recall some basic facts about Carnot groups, endpoint, and exponential maps.

Definition 2.1. A Carnot group is a nilpotent, connected, and simply connected Lie group $\mathbb{G}$ whose Lie algebra $\mathfrak{g}$ admits a stratification, i.e., a direct sum decomposition

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_s,$$

such that $[\mathfrak{g}_1, \mathfrak{g}_k] = \mathfrak{g}_{k+1}$ for $k = 1, \dots, s-1$ and $[\mathfrak{g}_1, \mathfrak{g}_s] = 0$.

The *left translation* by $g \in \mathbb{G}$ is the map $L_g \colon \mathbb{G} \to \mathbb{G}$ defined by $L_g(x) := gx$. Through this map, we have the identification

$$(L_g)_* \colon \ \mathfrak{g} \to T_g\mathbb{G}, \qquad \mathfrak{g} \ni X \mapsto (L_g)_* X \in T_g\mathbb{G}.$$

Every vector $X \in \mathfrak{g}$ defines the left-invariant vector field $X(g) := (L_g)_* X$. By abuse of notation, we identify the vector and the induced left-invariant vector field, in particular $X = X(1_{\mathbb{G}})$.

From now on, $I := [0,1] \subset \mathbb{R}$ denotes the unit interval. An absolutely continuous curve $\gamma\colon I \to \mathbb{G}$ is said to be *horizontal* if $\dot\gamma(t) \in (L_{\gamma(t)})_* \mathfrak{g}_1$ for a.e. $t \in I$, that is,

$$\left(L_{\gamma(t)}^{-1}\right)_* \dot\gamma(t) = Y(t) \qquad \text{a.e.},$$

for some function $Y \in L^\infty(I, \mathfrak{g}_1)$, called the *control* of γ.

Definition 2.2. The *endpoint map* is the mapping

$$E\colon\ L^\infty(I, \mathfrak{g}_1) \to \mathbb{G}, \qquad E(Y) := \gamma(1),$$

where γ is the unique solution to the Cauchy problem

$$\left(L_{\gamma(t)}^{-1}\right)_* \dot\gamma(t) = Y(t), \qquad t \in I,$$
$$\gamma(0) = 1_{\mathbb{G}}.$$

Given $X \in \mathfrak{g}_1$, the *endpoint map based at* X is the mapping $E_X\colon L^\infty(I, \mathfrak{g}_1) \to \mathbb{G}$ defined by $E_X(\cdot) := E(X + \cdot)$.

In the following, we need the following well known result, which is proved, for instance, in [1, Section 8].

Lemma 2.3. *For every $X \in \mathfrak{g}_1$, the endpoint map $E_X\colon L^\infty(I, \mathfrak{g}_1) \to \mathbb{G}$ is of class C^1 as soon as $L^\infty(I, \mathfrak{g}_1)$ is endowed with the L^2 topology.*

If $X \in \mathfrak{g}$ and $\gamma\colon I \to \mathbb{G}$ is the integral curve of (the left-invariant vector field identified with) X with $\gamma(0) = 1_{\mathbb{G}}$, then the *exponential map* is defined by

$$\exp\colon\ \mathfrak{g} \to \mathbb{G}, \qquad \exp(X) := \gamma(1).$$

Notice that $\gamma(t) = \exp(tX)$. Such a curve is called a *straight line*.

We denote the flow of a left-invariant vector field X by $\mathrm{e}^{tX}\colon \mathbb{G} \to \mathbb{G}$. We have that

$$\mathrm{e}^{tX}(g) = g \exp(tX).$$

For every $\lambda \in \mathbb{R}$, we define the dilation on the Lie algebra $\delta_\lambda\colon \mathfrak{g} \to \mathfrak{g}$ as the linear map such that

$$\delta_\lambda(X) = \lambda^i X \qquad \text{for all } X \in \mathfrak{g}_i.$$

By abuse of notation, we use the same symbol to denote the dilation on the group $\delta_\lambda\colon \mathbb{G} \to \mathbb{G}$ defined by

$$\delta_\lambda(\exp(X)) = \exp(\delta_\lambda(X)) \tag{2.1}$$

for every $X \in \mathfrak{g}$, so that $(\delta_\lambda)_* = \delta_\lambda$. In particular, for every $X \in \mathfrak{g}_1$ and $Y \in L^\infty(I, \mathfrak{g}_1)$, one has

$$\delta_\lambda(E_X(Y)) = E_{\lambda X}(\lambda Y). \tag{2.2}$$

3. Proof of main result

Let us give a slightly more general form of the definitions of (P) and (SP).

Definition 3.1. Let V be a vector subspace of $L^\infty(I, \mathfrak{g}_1)$. A vector $X \in \mathfrak{g}_1$ is *V-pliable* if the map E_X restricted to V is open at 0. $L^\infty(I, \mathfrak{g}_1)$-pliability coincides with pliability as defined by condition (P).

Definition 3.2. Let V be a vector subspace of $L^\infty(I, \mathfrak{g}_1)$. The vector $X \in \mathfrak{g}_1$ is said to be *strongly V-pliable* if for all $\eta > 0$ there exists $Y \in V$ such that

(i) $\|Y\|_\infty < \eta$;
(ii) $E_X(Y) = E_X(0)$;

(iii) $E_X|_V$ is a submersion at Y (i.e., the differential of $E_X|_V$ at Y is surjective).

$L^\infty(I,\mathfrak{g}_1)$-strong pliability coincides with strong pliability as defined by condition (SP).

Remark 3.3. In [4, Definition 3.4] and [11, Definition 4.1], the pliability and the strong pliability of a vector X are defined in terms of the openness and the submersion property of the extended map $\mathcal{E}_X(Y) := (E_X(Y), Y(1))$ restricted to the space

$$\mathcal{C}_0 := \big\{Y \in C^0(I,\mathfrak{g}_1) \mid Y(0) = 0\big\}.$$

However, in [4, Proposition 3.7] the openness of $\mathcal{E}_X|_{\mathcal{C}_0}$ is shown to be equivalent to the $\mathcal{C}_0$-pliability (in the sense of Definition 3.1), while in [11, Lemma 4.2] the strong pliability as defined in [11, Definition 4.1] is shown to be equivalent to that introduced in Definition 3.2. As we will see in Theorem 3.5 (considering both $V = \mathcal{C}_0$ and $V = L^\infty(I,\mathfrak{g}_1)$), it turns out that X is $\mathcal{C}_0$-pliable if and only if it is pliable, since both properties are equivalent to the p-free (H)-condition.

Before stating Theorem 1.1 in a slightly generalized version, let us introduce a useful notion.

Definition 3.4. We say that a subspace V of $L^\infty(I,\mathfrak{g}_1)$ is *L^∞-stably L^2-dense in* $L^\infty(I,\mathfrak{g}_1)$ if there exists a constant $C > 0$ such that, for every $Y \in L^\infty(I,\mathfrak{g}_1)$, there exists a sequence $(Z_k)_{k\in\mathbb{N}}$ in V converging to Y in $L^2(I,\mathfrak{g}_1)$ and such that

$$\|Z_k\|_\infty \le C\|Y\|_\infty \qquad \text{for every } k \in \mathbb{N}.$$

Examples of L^∞-stably L^2-dense subspaces of $L^\infty(I,\mathfrak{g}_1)$ include $\mathcal{C}_0$ and the subspace of $L^\infty(I,\mathfrak{g}_1)$ made of piecewise constant functions.

Theorem 1.1 is a corollary of the following result.

Theorem 3.5. *Let $\mathbb{G}$ be a Carnot group with Lie algebra $\mathfrak{g}$. Let V be an L^∞-stably L^2-dense subspace of $L^\infty(I,\mathfrak{g}_1)$. For every $X \in \mathfrak{g}_1$, the following conditions are equivalent:*

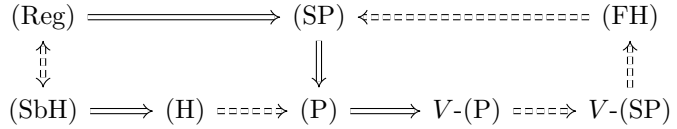
- V-(P) *X is V-pliable;*
- V-(SP) *X is strongly V-pliable;*
- (FH) *X satisfies the p-free* (H)*-condition.*

Moreover, X satisfies the submersive (H)*-condition* (SbH) *if and only if X is a regular point of E* ((Reg)*-condition*)*, and both conditions imply*

- (H) *X satisfies the* (H)*-condition;*

which, in turn, implies the three equivalent conditions V-(P)*, V-*(SP)*, and* (FH)*.*

The structure of the proof of Theorem 3.5 is summarized in the following diagram:



Here solid arrows denote implications that follow immediately from the definitions, while dashed arrows are proved via different lemmas.

The implication V-(P) ⇢ V-(SP) is a consequence of Lemmas 3.6 and 3.8 (as explained in Remark 3.9). The implications (H) ⇢ (P) and (FH) ⇢ (SP) are proved in Lemma 3.10. The implication V-(SP) ⇢ (FH) is proved in Lemma 3.11. The equivalence (Reg) ⇿ (SbH) is proved in Lemma 3.12. Finally, we notice that the implications (Reg) ⟹ (SP) and (SbH) ⟹ (FH) cannot be reversed, see Remark 3.13.

Lemma 3.6. *A vector $X \in \mathfrak{g}_1$ is pliable if and only if it is strongly pliable.*

Proof. The fact that strong pliability implies pliability is trivial. The converse can be proved in the following way.

Assume that $X \in \mathfrak{g}_1$ is pliable. It results from the homogeneity property (2.2) of E_X that $\frac{X}{2}$ and $-\frac{X}{2}$ are also pliable. Let $g_1 = \exp(X)$ and $g_{\frac{1}{2}} = \exp\left(\frac{X}{2}\right)$. Fix $\eta > 0$ and denote by B_η be the ball of radius η in $L^\infty(I, \mathfrak{g}_1)$ centered at 0. By pliability of the vector $-\frac{X}{2}$, the set $\mathcal{U}_- = g_1 E_{-\frac{X}{2}}(B_\eta)$ is a neighborhood of $g_{\frac{1}{2}}$. Now we can fix $0 < \eta' < \eta$ such that $\mathcal{U} := E_{\frac{X}{2}}(B_{\eta'}) \subset \mathcal{U}_-$. Since $\frac{X}{2}$ is pliable, $\mathcal{U}$ is a neighborhood of $g_{\frac{1}{2}}$.

Consider

$$\left(L^{-1}_{\gamma(s)}\right)_* \dot\gamma(s) = \frac{X(\gamma(s))}{2} + Y_s(\gamma(s)),$$

which can be seen as a control system with control $s \mapsto Y_s$ taking values in the ball of center 0 and radius η' in $\mathfrak{g}_1$. By a well-known result in geometric control theory (see, e.g., [5, Section 3, Theorem 1]), there exist $k \in \mathbb{N}$, $Y_1, \dots, Y_k \in \mathfrak{g}_1$ with $\|Y_i\| < \eta'$ for $i = 1, \dots, k$, and $t_1, \dots, t_k > 0$ such that $(s_1, \dots, s_k) \mapsto \mathrm{e}^{s_k(\frac{X}{2}+Y_k)} \circ \dots \circ \mathrm{e}^{s_1(\frac{X}{2}+Y_1)}(1_{\mathbb{G}})$ is a submersion at $(t_1, \dots, t_k)$, with $\sum_{i=1}^k t_i$ arbitrary small. Since t_k can be increased without affecting the submersion property, we can assume without loss of generality that $\sum_{i=1}^k t_i = 1$.

Set, for every $\varepsilon \in \mathbb{R}$ and $j \in \{1, \dots, k\}$,

$$\tilde Y_j(\varepsilon) = Y_j + \frac{\varepsilon}{t_j}\left(\frac{X}{2} + Y_j\right),$$

and notice that

$$t_j\left(\frac{X}{2} + \tilde Y_j(\varepsilon)\right) = (t_j + \varepsilon)\left(\frac{X}{2} + Y_j\right), \qquad j \in \{1, \dots, k\},\ \varepsilon \in \mathbb{R}.$$

As a consequence, the map

$$(\varepsilon_1, \dots, \varepsilon_k) \mapsto \mathrm{e}^{t_k(\frac{X}{2}+\tilde Y_k(\varepsilon_k))} \circ \dots \circ \mathrm{e}^{t_1(\frac{X}{2}+\tilde Y_1(\varepsilon_1))}(1_{\mathbb{G}})$$

is a submersion at $(\varepsilon_1, \dots, \varepsilon_k) = (0, \dots, 0)$. In particular, letting $Y' \in L^\infty(I, \mathfrak{g}_1)$ be equal to Y_1 on $[0, t_1)$, then to Y_2 on $[t_1, t_1 + t_2)$, and so on, we have that $Y' \in B_{\eta'}$ and $E_{\frac{X}{2}}$ is a submersion at Y'.

Letting $g'_{\frac{1}{2}} = E_{\frac{X}{2}}(Y') \in \mathcal{U}$, there exists $Y'' \in B_\eta$ such that $g_1 E_{-\frac{X}{2}}(Y'') = g'_{\frac{1}{2}}$. We set

$$Y(t) = \begin{cases} 2Y'(2t), & t \in \left[0, \frac{1}{2}\right], \\ -2Y''(2(1-t)), & t \in \left[\frac{1}{2}, 1\right]. \end{cases}$$

Then, we have $E_X(Y) = g_1$ and, since $\mathrm{d}_{Y'} E_{\frac{X}{2}}$ is surjective, also the differential at $\mathrm{d}_Y E_X$ is surjective. Since $\|Y\|_\infty < 2\eta$ and η is arbitrarily small, the strong pliability of X is proved. □

Remark 3.7. Let us stress that Lemma 3.6 relies essentially on the special properties of the endpoint map, in the sense that for the general case of a smooth open map F at a point X, it is not true that there exist points arbitrarily close to X whose image through F coincide with $F(X)$ and at which F is a submersion. Consider, for instance, $F\colon L^\infty(I, \mathbb{R}) \ni X \mapsto \left(\int_I X(t)\mathrm{d}t\right)^3 \in \mathbb{R}$. The map F is open at 0, but for every $X \in L^\infty(I, \mathbb{R})$ with $F(X) = 0$ the differential of F at X is the null map.

Lemma 3.8. *Assume that V is an L^∞-stably L^2-dense subspace of $L^\infty(I, \mathfrak{g}_1)$. Then, a vector $X \in \mathfrak{g}_1$ is strongly pliable if and only if X is strongly V-pliable.*

Proof. Since one implication is trivial, let us assume that X is strongly pliable and let us prove that it is strongly V-pliable. Fix $\eta > 0$ and $C > 0$ as in Definition 3.4. Then there exists $Y \in L^\infty(I, \mathfrak{g}_1)$ with $\|Y\|_\infty < \frac{\eta}{2C}$ such that $E_X(Y) = E_X(0)$ and $\mathrm{d}E_X(Y)$ is surjective. Pick $Y_1, \dots, Y_d \in L^\infty(I, \mathfrak{g}_1)$ such that $\mathrm{d}E_X(Y)Y_1, \dots, \mathrm{d}E_X(Y)Y_d$ is a basis of $T_{E_X(0)}\mathbb{G}$, where d is the dimension of $\mathbb{G}$. Since V is dense in $L^\infty(I, \mathfrak{g}_1)$ for the L^2-topology, and the map $Z \mapsto \mathrm{d}E_X(Y)Z$ is continuous in $L^2(I, \mathfrak{g}_1)$, we can assume that $Y_1, \dots, Y_d \in V$ by a standard approximation argument.

Consider the function $\mathbb{R}^d \ni (\alpha_1, \dots, \alpha_d) \mapsto E_X(Y + \alpha_1 Y_1 + \cdots + \alpha_d Y_d)$. By the inverse function theorem, for $r > 0$ sufficiently small, such a function diffeomorphically maps $D = \{(\alpha_1, \dots, \alpha_d) \in \mathbb{R}^d \mid |\alpha_1|, \dots, |\alpha_d| < r\}$ into $E_X(D)$. Without loss of generality,

$$r < \frac{\eta}{2(\|Y_1\|_\infty + \cdots + \|Y_d\|_\infty)}.$$

Let $Z \in V$ be such that $\|Z - Y\|_2 < \varepsilon$, for some $\varepsilon > 0$ to be fixed later. Since V is L^∞-stably L^2-dense in $L^\infty(I, \mathfrak{g}_1)$, we can assume that $\|Z\|_\infty \le C\|Y\|_\infty < \frac{\eta}{2}$. Notice that $E_X(Z + \alpha_1 Y_1 + \cdots + \alpha_d Y_d)$ is arbitrarily close to $E_X(Y + \alpha_1 Y_1 + \cdots + \alpha_d Y_d)$ as ε goes to zero, uniformly with respect to $(\alpha_1, \dots, \alpha_d) \in D$ in any fixed system of coordinates. Using a classical degree argument similar to the one in [11, Lemma 5.5], and choosing $\varepsilon > 0$ small enough, we have that $E_X(0) = E_X(w)$ for some $w \in \{Z + \alpha_1 Y_1 + \cdots + \alpha_d Y_d \mid |\alpha_1|, \dots, |\alpha_d| < r\} \subset V$ and $\mathrm{d}E_X(w)Y_1, \dots, \mathrm{d}E_X(w)Y_d$ is a basis of $T_{E_X(0)}\mathbb{G}$.

Notice that, up to further reducing ε if necessary,

$$\|w\|_\infty \le \|Z\|_\infty + r(\|Y_1\|_\infty + \cdots + \|Y_d\|_\infty) < \eta.$$

Therefore, Properties (i), (ii), and (iii) in Definition 3.2 are satisfied by $w \in V$. □

Remark 3.9. Let V be an L^∞-stably L^2-dense subspace of $L^\infty(I, \mathfrak{g}_1)$. Then, by the two previous results, it follows that

$$\text{str. } V\text{-pliable} \implies V\text{-pliable} \implies \text{pliable} \overset{\text{Lemma 3.6}}{\implies} \text{str. pliable} \overset{\text{Lemma 3.8}}{\implies} \text{str. } V\text{-pliable}.$$

Lemma 3.10. *If a vector $X \in \mathfrak{g}_1$ satisfies the* (H)*-condition* (*respectively, the* (FH)*-condition*)*, then it is pliable* (*respectively, strongly pliable*)*.*

Proof. If $X \in \mathfrak{g}_1$ satisfies the (H)-condition, there exists $p \in \mathbb{N}$ such that the map

$$(Y_1, \dots, Y_p) \mapsto \exp(Y_1) \cdots \exp(Y_p)$$

is open at $(X, \dots, X)$. Hence, given any $\eta > 0$, $\big\{\mathrm{e}^{Y_p} \circ \cdots \circ \mathrm{e}^{Y_1}(1_\mathbb{G}) \mid \|Y_i - X\| < \eta \text{ for } i = 1, \dots, p\big\}$ is a neighborhood of $\mathrm{e}^{pX}(1_\mathbb{G})$. Let $\delta_{\frac{1}{p}}$ be the dilation in $\mathbb{G}$ of parameter $1/p$. Then, thanks to (2.1), for every $(Y_1, \dots, Y_p) \in (\mathfrak{g}_1)^p$ we have

$$\mathrm{e}^{\frac{1}{p}Y_p} \circ \cdots \circ \mathrm{e}^{\frac{1}{p}Y_1}(1_\mathbb{G}) = \delta_{\frac{1}{p}}\big(\mathrm{e}^{Y_p} \circ \cdots \circ \mathrm{e}^{Y_1}(1_\mathbb{G})\big).$$

Since, moreover, $\mathrm{e}^{\frac{1}{p}Y_p} \circ \cdots \circ \mathrm{e}^{\frac{1}{p}Y_1}(1_\mathbb{G})$ is the image by E_X of the control constantly equal to Y_i on $((i-1)/p, i/p)$ for $i = 1, \dots, p$, it turns out that the image by E_X of the set

$$\big\{Y \in L^\infty(I, \mathfrak{g}_1) \mid \|Y\|_\infty < \eta,\ Y|_{(i/p,(i+1)/p)} \text{ constant } \forall i = 0, \dots, p-1\big\}$$

is a neighborhood of $\mathrm{e}^X(1_\mathbb{G})$, which implies that X is pliable.

The proof in the case of the (FH)-condition is similar. □

Lemma 3.11. *If $X \in \mathfrak{g}_1$ is strongly pliable, then it satisfies the* (FH)*-condition.*

Proof. Let $X \in \mathfrak{g}_1$ be strongly pliable, and denote by $\mathrm{PC} = \mathrm{PC}(I, \mathfrak{g}_1)$ the subspace of $L^\infty(I, \mathfrak{g}_1)$ made of piecewise constant functions with rational discontinuity points. By Remark 3.9, X is strongly PC-pliable.

Fix $\eta > 0$. By the strong PC-pliability, there exist $Z, Y_1, \dots, Y_d \in \mathrm{PC}$, with $\|Z\|_\infty < \eta$, such that $E_X(Z) = E_X(0)$ and $\mathrm{d}E_X(Z)Y_1, \dots, \mathrm{d}E_X(Z)Y_d$ is a basis of $T_{E_X(0)}\mathbb{G}$. By definition of the space PC, there exists an integer $p \geq 2$ such that $Z|_{((i-1)/p,i/p)}, Y_1|_{((i-1)/p,i/p)}, \dots, Y_d|_{((i-1)/p,i/p)}$ are constant for every $i = 1, \dots, p$.

Denote by Z_i and $Y_{j,i}$ the values of Z and Y_j on the interval $\left(\frac{i-1}{p}, \frac{i}{p}\right)$. Then the map $\Gamma^{(p)} \colon (\mathfrak{g}_1)^p \to \mathbb{G}$ defined by

$$\Gamma^{(p)}(W_1, \dots, W_p) := \mathrm{e}^{W_p} \circ \cdots \circ \mathrm{e}^{W_1}(1_\mathbb{G}),$$

satisfies $\Gamma^{(p)}(X + Z_1, \dots, X + Z_p) = \Gamma^{(p)}(X, \dots, X)$. It remains to show that $\Gamma^{(p)}$ is a submersion at $(X + Z_1, \dots, X + Z_p)$.

We have that $Y_j = \sum_{i=1}^p Y_{j,i}\chi_{((i-1)/p,i/p)}$ and so

$$\mathrm{d}E_X(Z)Y_j = \sum_{i=1}^p \mathrm{d}E_X(Z)Y_{j,i}\chi_{(\frac{i-1}{p},\frac{i}{p})}.$$

Moreover,

$$\begin{aligned}
&\mathrm{d}E_X(Z)Y_{j,i}\chi_{(\frac{i-1}{p},\frac{i}{p})} \\
&= \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\bigg|_{\varepsilon=0} \left(\mathrm{e}^{\frac{X+Z_p}{p}} \circ \cdots \circ \mathrm{e}^{\frac{X+Z_{i+1}}{p}} \circ \mathrm{e}^{\frac{X+Z_i+\varepsilon Y_{j,i}}{p}} \circ \mathrm{e}^{\frac{X+Z_{i-1}}{p}} \circ \cdots \circ \mathrm{e}^{\frac{X+Z_1}{p}}(1_\mathbb{G})\right) \\
&\overset{(2.1)}{=} \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\bigg|_{\varepsilon=0} \delta_{\frac{1}{p}}\left(\mathrm{e}^{X+Z_p} \circ \cdots \circ \mathrm{e}^{X+Z_{i+1}} \circ \mathrm{e}^{X+Z_i+\varepsilon Y_{j,i}} \circ \mathrm{e}^{X+Z_{i-1}} \circ \cdots \circ \mathrm{e}^{X+Z_1}(1_\mathbb{G})\right) \\
&= \left(\delta_{\frac{1}{p}}\right)_* \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\bigg|_{\varepsilon=0} \left(\mathrm{e}^{X+Z_p} \circ \cdots \circ \mathrm{e}^{X+Z_{i+1}} \circ \mathrm{e}^{X+Z_i+\varepsilon Y_{j,i}} \circ \mathrm{e}^{X+Z_{i-1}} \circ \cdots \circ \mathrm{e}^{X+Z_1}(1_\mathbb{G})\right) \\
&= \left(\delta_{\frac{1}{p}}\right)_* \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\bigg|_{\varepsilon=0} \Gamma^{(p)}(X + Z_1, \dots, X + Z_{i-1}, X + Z_i + \varepsilon Y_{j,i}, X + Z_{i+1}, \dots, X + Z_p) \\
&= \left(\delta_{\frac{1}{p}}\right)_* \mathrm{d}\Gamma^{(p)}(X + Z_1, \dots, X + Z_p) \cdot (0, \dots, 0, Y_{j,i}, 0, \dots, 0). \qquad (3.1)
\end{aligned}$$

So the image of the differential of $\Gamma^{(p)}$ at $(X+Z_1, \dots, X+Z_p)$ contains the image by $(\delta_p)_* = \left(\delta_{\frac{1}{p}}\right)_*^{-1}$ of the span of $\mathrm{d}E_X(Z)Y_1, \dots, \mathrm{d}E_X(Z)Y_d$. Hence $\Gamma^{(p)}$ is a submersion at $(X+Z_1, \dots, X+Z_p)$. □

Lemma 3.12. *A vector $X \in \mathfrak{g}_1$ satisfies the submersive* (H)*-condition* (SbH) *if and only if it is a regular point of the endpoint map E.*

Proof. On the one hand, if $X \in \mathfrak{g}_1$ satisfies (SbH), then a computation analogous to (3.1) with $Z = 0$ shows that $\mathrm{d}E_X(0)$ is onto. On the other hand, note that the argument of Lemma 3.8 shows that, if E_X is a submersion at 0, then E_X restricted to PC is also a submersion at 0. Then the argument used in the proof of Lemma 3.11 shows that X satisfies the submersive (H)-condition. □

Remark 3.13. The (H)-condition is strictly weaker than the submersive (H)-condition (SbH). In particular, also pliability, strong pliability, and the (FH)-condition are strictly weaker than (SbH). Indeed, Carnot groups of step two are known to satisfy the (H)-condition [8, Theorem 2.1], while non-Métivier step two Carnot groups admit abnormal curves of the form $t \mapsto \exp(tX)$ (see [6, Proposition 3.6]) and, hence, vectors X that do not satisfy the submersive (H)-condition.

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